

26.4 Collinearity and prediction

By Theorem 26.1.1(d), collinearity leaves the total variance of the fitted values, the predictions at the design points, untouched. At a new point $\mathbf{x}_0$, collinearity damages a prediction exactly to the extent that the point asks for information along a weak direction. Example (Predicting along and across the pattern) showed this contrast on simulated data; this section proves it in general and measures it on real data.

26.4.1 The variance of a prediction

Keep the notation of Section 26.1: $\mathbf{X}^\top \mathbf{X} = \sum_{\ell} \lambda_{\ell} \mathbf{v}_{\ell} \mathbf{v}_{\ell}^\top$, and let

$$m_{\ell} = \frac{\lambda_{\ell}}{n} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_{(i)}^\top \mathbf{v}_{\ell})^2$$

be the mean square of the design rows along the direction $\mathbf{v}_{\ell}$. A small m_{ℓ} means that the rows hardly vary along $\mathbf{v}_{\ell}$: the data cloud is thin in that direction. The estimate of the mean response at $\mathbf{x}_0$ is $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$.

Proposition 26.4.1. Prediction under collinearity

Let $\mathbf{X}$ have rank p and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$.

- (a) $\text{Var}(\hat{Y}_0) = \frac{\sigma^2}{n} \sum_{\ell=1}^p \frac{(\mathbf{x}_0^\top \mathbf{v}_{\ell})^2}{m_{\ell}}$.
- (b) If $\mathbf{x}_0 = \mathbf{X}^\top \mathbf{w}$ for some $\mathbf{w} \in \mathbb{R}^n$, then $\text{Var}(\hat{Y}_0) = \sigma^2 \|\mathbf{M}\mathbf{w}\|^2 \leq \sigma^2 \|\mathbf{w}\|^2$. If $\mathbf{x}_0$ is a convex combination of the rows of $\mathbf{X}$, then $\text{Var}(\hat{Y}_0) \leq \sigma^2 \max_i h_{ii} \leq \sigma^2$. Neither bound involves the eigenvalues.
- (c) Let $\mathbf{x}_0$ be random, independent of $\mathbf{Y}$, with $\text{E}(\mathbf{x}_0 \mathbf{x}_0^\top) = \boldsymbol{\Sigma}_0$. Then

$$\text{E}\{\text{Var}(\hat{Y}_0 \mid \mathbf{x}_0)\} = \frac{\sigma^2}{n} \sum_{\ell=1}^p \frac{\mathbf{v}_{\ell}^\top \boldsymbol{\Sigma}_0 \mathbf{v}_{\ell}}{m_{\ell}}.$$

If $\mathbf{x}_0$ is drawn uniformly from the rows of $\mathbf{X}$, this equals $\sigma^2 p/n$, whatever the eigenvalues. For every $\boldsymbol{\Sigma}_0$ it is at least $(\sigma^2/n) \mathbf{v}_p^\top \boldsymbol{\Sigma}_0 \mathbf{v}_p / m_p$.

Proof.

- (a) This is Proposition 19.6.1(a) with $\mathbf{a} = \mathbf{x}_0$ and $\lambda_{\ell} = nm_{\ell}$. The second expression for m_{ℓ} is $\|\mathbf{X}\mathbf{v}_{\ell}\|^2/n$.
- (b) By Theorem 6.3.6,

$$\text{Var}(\hat{Y}_0) = \sigma^2 \mathbf{w}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{w} = \sigma^2 \mathbf{w}^\top \mathbf{M}\mathbf{w} = \sigma^2 \|\mathbf{M}\mathbf{w}\|^2,$$

and $\|\mathbf{M}\mathbf{w}\| \leq \|\mathbf{w}\|$ (Proposition 6.3.4). If $\mathbf{x}_0 = \sum_i w_i \mathbf{x}_{(i)}$ with $w_i \geq 0$ and $\sum_i w_i = 1$, then $\mathbf{x}_0 = \mathbf{X}^\top \mathbf{w}$, and by the triangle inequality and Proposition 6.8.1(a),

$$\|\mathbf{M}\mathbf{w}\| \leq \sum_i w_i \|\mathbf{M}\mathbf{e}_i\| = \sum_i w_i \sqrt{h_{ii}} \leq \max_i \sqrt{h_{ii}},$$

and $h_{ii} \leq 1$.

- (c) Conditionally on $\mathbf{x}_0$, (a) holds. Taking expectations, $E(\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0) = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \Sigma_0) = \sum_{\ell} \mathbf{v}_{\ell}^\top \Sigma_0 \mathbf{v}_{\ell} / \lambda_{\ell}$. If $\mathbf{x}_0$ is uniform on the rows, $\Sigma_0 = \mathbf{X}^\top \mathbf{X} / n$, and the trace is $\text{tr}(\mathbf{I}_p) / n = p / n$. The lower bound keeps the term $\ell = p$, since all terms are nonnegative. ■

Part (b) answers the first question of Exercise (C1): inside the convex hull of the data, a prediction is never worse than the worst fitted value, however collinear the regressors. Part (a) says what “the pattern of the data” means: a point outside the hull is still predicted well if its coordinates along the weak directions are small compared with $\sqrt{m_{\ell}}$, and a point inside the range of every regressor separately can be far off the pattern (Example (Hidden extrapolation)). By part (c), if future regressors are generated the way the design was, the average prediction variance is $\sigma^2 p / n$; if a relation between the regressors breaks down, the term $\mathbf{v}_p^\top \Sigma_0 \mathbf{v}_p / m_p$ can be enormous.

Warning (Variance is not the whole risk). [Proposition 26.4.1](#) is about variance under a correct model. Off the pattern of the data the model itself has not been checked, so its bias there is unknown ([Section 12.3](#)). A small prediction variance is not a guarantee.

Example. Predicting consumption along and across the pattern

In the consumption regression of Example (Large factors, precise estimate), $s = 58.1$ (billions of 2005 dollars). At 1985Q1, a design point with population 237.9 million, the estimated mean has standard deviation 8.5, below the design average $s\sqrt{p/n} = 10.8$. Midway between the first quarters of 1969 and 1999 the standard deviation is 7.8, and by [Proposition 26.4.1\(b\)](#) no point in the convex hull of the data can exceed $s\sqrt{\max_i h_{ii}} = 23.3$.

Now keep every regressor at its 1985 value except population, which is lowered by ten per cent, to 214.1 million. Population did take that value, in 1974, and income took its 1985 value, but never together. The standard deviation of the estimated mean becomes 36.8, 4.3 times as large. Decomposing the variance over the canonical directions of the scaled design, as in the proof of Theorem 26.3.2, the direction with condition index 270.9, the income–population near-dependency of Example (A consumption function), contributes 93.9 per cent of the variance at the new point and 0.2 per cent at the 1985 quarter itself.

[Figure 26.4.1](#) shows 2000 parametric replicates: responses generated from the fitted model with normal errors of standard deviation s , each refitted. The income and population coefficients are strongly negatively correlated: their exact correlation is -0.842 (-0.832 in the replicates). The prediction at 1985Q1 barely moves while the one off the pattern spreads widely.

```
import numpy as np
import statsmodels.api as sm
macro = sm.datasets.macrodta.load_pandas().data
names = ["realdpi", "pop", "cpi", "m1", "tbilrate", "unemp"]
X = sm.add_constant(macro[names].to_numpy())
y = macro["realcons"].to_numpy()
n, p = X.shape
fit = sm.OLS(y, X).fit()
s = np.sqrt(fit.scale)
G = np.linalg.inv(X.T @ X) # estimate of sigma
```

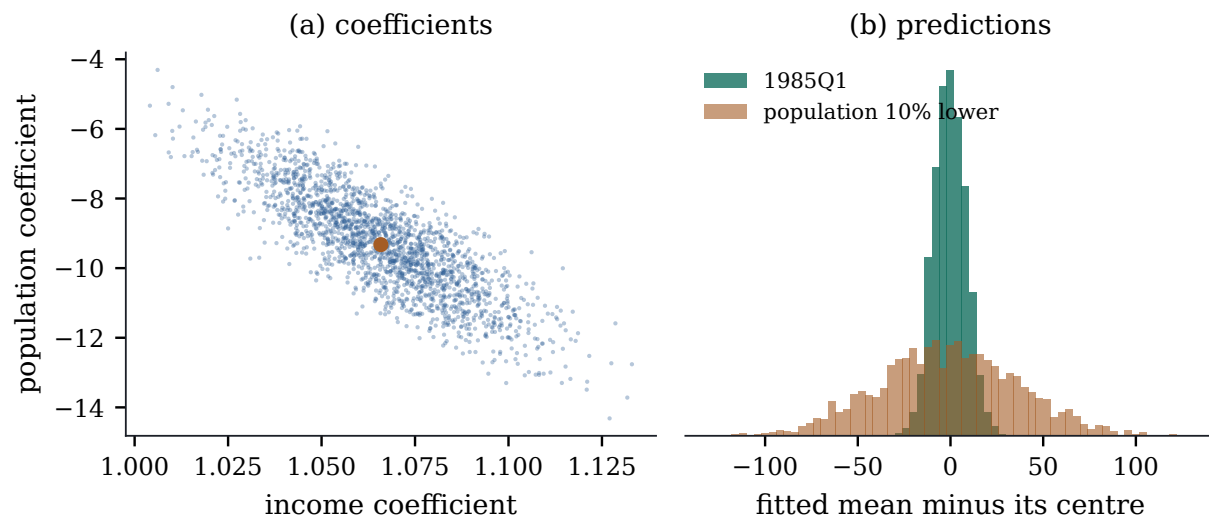


Figure 26.4.1: Parametric replicates for [Example \(Predicting consumption along and across the pattern\)](#). (a) Income and population coefficients. (b) Estimated mean, centred, at 1985Q1 and at the same point with population ten per cent lower.

```
x_on = X[104]                                # 1985Q1, a design point
x_mix = 0.5 * (X[40] + X[160])                # midway between 1969Q1 and
1999Q1
x_off = x_on.copy()
x_off[2] *= 0.90                             # 1985Q1 with population 10%
lower
for label, x0 in [("1985Q1", x_on), ("midway", x_mix), ("off pattern", x_off)]:
    print(f"{label:13s} fitted {x0 @ fit.params:8.1f}    sd {s * np.sqrt(x0 @ G @ x0)
:6.1f}")
```

[Run this code online](#)

```
d = np.linalg.norm(X, axis=0)
U, mu, Vt = np.linalg.svd(X / d, full_matrices=False) # scaled design, as in Section
26.3

def shares(x0):
    """Contributions of the canonical directions to Var(x0' beta_hat), as fractions.
    """
    c = (Vt @ (x0 / d)) ** 2 / mu**2                # (x0_scaled' v_k)^2 / mu_k^2
    return c / c.sum()

np.set_printoptions(precision=3, suppress=True)
print("condition indices:", mu[0] / mu)
print("shares, 1985Q1      :", shares(x_on))
print("shares, off pattern :", shares(x_off))
```

[Run this code online](#)

```
rng = np.random.default_rng(2604)
reps = 2000
Ystar = X @ fit.params + s * rng.normal(size=(reps, n)) # parametric replicates
, same design
Bstar = np.linalg.solve(X.T @ X, X.T @ Ystar.T).T
```

```
print("sd of income, population coefficients:", Bstar[:, 1:3].std(axis=0))
print("correlation of the two:", np.corrcoef(Bstar[:, 1], Bstar[:, 2])[0, 1])
print("sd of predictions, 1985Q1 / off:", (Bstar @ x_on).std(), (Bstar @ x_off).std())
```

[Run this code online](#)

26.4.2 Coefficients against predictions

A coefficient β_j is the difference between the mean responses at two points that differ by one unit in x_j and agree in every other regressor; with collinear data at least one of them lies off the pattern. A regression built for forecasting, where the relations among the regressors are expected to persist, can ignore collinearity; one built to attribute effects to individual regressors cannot.

26.4.3 Exercises

26.4.3.1 A. Check your understanding

Exercise. A1

For the design of Example (Two regressors that move together), rank the following new points by the variance of the fitted mean, without computing: the centroid; $(x_1, x_2) = (3, 3)$; $(x_1, x_2) = (0.5, -0.5)$; the average of the two design points with the largest x_1 . Which bound of [Proposition 26.4.1](#) applies to each? Compare [Figure 19.6.1](#).

26.4.3.2 B. Practice

Exercise. B1

Show that the bound $\text{Var}(\hat{Y}_0) \leq \sigma^2 \|\mathbf{w}\|^2$ of [Proposition 26.4.1\(b\)](#) is attained iff $\mathbf{w} \in \mathcal{C}(\mathbf{X})$, and that $\min\{\|\mathbf{w}\|^2 : \mathbf{X}^\top \mathbf{w} = \mathbf{x}_0\}$ equals $\text{Var}(\hat{Y}_0)/\sigma^2$. Compare with [Proposition 12.3.2\(a\)](#).

Solution. $\|\mathbf{M}\mathbf{w}\|^2 = \|\mathbf{w}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{w}\|^2$, so equality holds iff $(\mathbf{I} - \mathbf{M})\mathbf{w} = \mathbf{0}$, that is, $\mathbf{w} \in \mathcal{C}(\mathbf{X})$. Every $\mathbf{w}$ with $\mathbf{X}^\top \mathbf{w} = \mathbf{x}_0$ has the same $\mathbf{M}\mathbf{w}$, because two such vectors differ by an element of $\mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$. The minimum norm is therefore attained at $\mathbf{w} = \mathbf{M}\mathbf{w}$ and equals $\|\mathbf{M}\mathbf{w}\|^2 = \text{Var}(\hat{Y}_0)/\sigma^2$. This is [Proposition 12.3.2\(a\)](#) with $h_0 = \text{Var}(\hat{Y}_0)/\sigma^2$: the leverage of a new point is the smallest squared norm of a set of weights that builds $\mathbf{x}_0$ from the rows of the design.

Exercise. B2

Let the model have an intercept and centred regressors, so that the rows of $\mathbf{X}$ are $(1, \mathbf{z}_{(i)}^\top)$ with $\sum_i \mathbf{z}_{(i)} = \mathbf{0}$. For a design row i and $t > 1$, let $\mathbf{x}_0 = (1, t\mathbf{z}_{(i)}^\top)^\top$, an extrapolation from the centroid along the ray through a design point. Show that $\text{Var}(\hat{Y}_0) = \sigma^2(1/n + t^2(h_{ii} - 1/n))$. Why does collinearity not enter, although $\mathbf{x}_0$ is outside the convex hull of the data when t is large?

Solution. With centred regressors $\mathbf{X}^\top \mathbf{X} = \text{diag}(n, \mathbf{Z}^\top \mathbf{Z})$, where $\mathbf{Z}$ has rows $\mathbf{z}_{(i)}^\top$, so $\text{Var}(\hat{Y}_0) = \sigma^2(1/n + t^2 \mathbf{z}_{(i)}^\top (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{z}_{(i)})$, and $h_{ii} = 1/n + \mathbf{z}_{(i)}^\top (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{z}_{(i)}$ by Proposition 6.8.2. The slope part of $\mathbf{x}_0$ is t times that of a row, so its coordinate along every eigenvector is t times the row's, and in each term of Proposition 26.4.1(a) it is penalized relative to the data exactly as much as the row is, times t^2 . Collinearity could enter only through a component along a weak direction that is large compared with the spread of the data there, and the row itself has no such component.

26.4.3.3 C. Going deeper

Exercise. C1

In the design of Example (Two regressors that move together), suppose future regressor values satisfy $x_2 - x_1 = \tau\zeta$ with ζ independent of x_1 , $E\zeta = 0$, $\text{Var}\zeta = 1$, and x_1 distributed as in the sample. Using Proposition 26.4.1(c), show that the average prediction variance is approximately $(\sigma^2/n)(2 + \tau^2/(2m_2))$, where $m_2 = \lambda_2/n$ is the mean square of the design along the weak slope direction. Check that this is $\sigma^2 p/n$ when τ^2 equals the within-sample mean square of $x_2 - x_1$, and discuss what happens when τ is several times larger. What would a sensible forecaster do?